

Fully funded PhD position in AI-based prediction and climate change attribution of extreme events

Application closing date: 31 December 2026 or until filled

Start date: Spring 2027 or as soon as possible thereafter

Duration: 4 years, full-time

Institution: Geosciences Institute (IGEO), National Research Council – Complutense University of Madrid (CSIC–UCM). Madrid, Spain

Project: ADAPT-X – AI-Driven AnticiPated aTtribution of eXtreme events

Supervision: Dr. Bernat Jiménez Esteve and Dr. David Barriopedro (co-supervisor)

Call

The [Instituto de Geociencias](#) (IGEO, CSIC–UCM) invites applications for a **fully funded 4-year PhD position** within **ADAPT-X** (AI-Driven AnticiPated aTtribution of eXtreme events) project (ref. 2025-T1/ECO-36122), funded by **Comunidad de Madrid** through the [César Nombela Talent Attraction Programme](#).

Recent AI weather prediction models can produce skillful global weather forecasts at a fraction of the computational cost of conventional numerical weather prediction models. ADAPT-X aims to exploit these new capabilities to understand the predictability of extreme events and develop methods capable of quantifying the influence of climate change before or as an event unfolds. The project combines atmospheric dynamics, weather forecasting, climate-change attribution, numerical modelling and artificial intelligence.

The PhD candidate will investigate the use of **AI-based weather prediction models for the prediction and climate-change attribution of extreme events** such as heatwaves, extreme precipitation, droughts and severe storms.

The research is expected to:

- Evaluate the predictive skill of AI weather models and how these models represent extreme events and the underlying dynamics.
- Build factual/counterfactual forecast experiments by initializing AI models under present-day and pre-industrial climate conditions.
- Check the physical consistency of the attributed response with dynamical and thermodynamic diagnostics.
- Benchmark AI-based attribution against physics-based modelling, including high-resolution NWP simulations for selected events.

The exact emphasis of the PhD thesis will be adapted to the candidate's background and interests. The project provides access to substantial computational resources, including GPU infrastructure for running physical and AI weather models, as well as technical support at IGEO-CSIC.

The successful candidate will join the [Physics of Climate and Climate Change](#) group of CSIC at IGEO, and the [STREAM](#) research group of UCM, which have long-standing experience in the fields of extreme events, atmospheric dynamics and extreme event attribution, including some recent developments in AI.

The student will be enrolled in the PhD Programme in Physics at UCM. The project includes opportunities for international collaboration and research stays with partner groups and institutions, as well as participation in international conferences and scientific publications.

Candidate profile

Applicants should hold, or be close to completing, a **Master's degree providing access to a PhD** programme. Suitable backgrounds include: Atmospheric sciences, meteorology, climate science, physics, applied mathematics, environmental sciences, computer science, data science, or related quantitative disciplines.

Candidates **should have**:

- Good scientific programming skills, particularly in Python.
- A strong quantitative background and interest in atmospheric and climate science.
- Good written and spoken English.
- Ability to work both independently and collaboratively.

Previous experience in one or more of the following areas would be advantageous, but is not required in all of them:

- Atmospheric dynamics and extreme events.
- Numerical weather or climate modelling.
- AI or machine-learning applications in weather and climate.
- Linux/UNIX environments, shell scripting, and high-performance computing.
- Working with large meteorological or climate datasets.

Employment conditions

The position is a **full-time 4-year** predoctoral employment contract based at IGEO in Madrid.

Salary: approximately €1,655 net per month, depending on personal tax circumstances, with full Spanish Social Security coverage and standard employee benefits.

How to apply

Applicants should send an email to Bernat Jiménez (bernatji@ucm.es), with the title "ADAPT-X PhD application – [Candidate name]". Applications should include a single PDF file with your name, including:

1. Curriculum Vitae, detailing relevant research, programming experience and publications, if applicable.
2. Motivation letter of approximately one page describing the candidate's background, research interests and motivation for undertaking this PhD.
3. Bachelor's and Master's academic transcripts.
4. Names and contact details of two potential referees.

Applications should be submitted by 31 December 2026, though submissions will be accepted after this date until the position is filled.

The Institution

[CSIC](#) (Spanish National Research Council) is the largest public research institution in Spain and the third one in Europe. It plays a key role in scientific and technological policy in Spain by carrying out research in all fields of knowledge. CSIC produces ~20% of the national scientific output and is listed

the 1st organization in Spain, the 3rd in H2020 (917 projects) and the 2nd in Horizon Europe (612 projects). It is also a main actor in the ERC programme (161 projects signed as Host Institution), the EIT's, KIC's, and International Programmes like LIFE and INTERREG. CSIC coordinates key infrastructures (ESFRI, EOSC), promotes Open Science, and holds the HR Excellence in Research award.

The successful candidate will join the [Instituto de Geociencias](#) (IGEO, Madrid), a Joint Research Center of CSIC and UCM. IGEO gathers employees and resources from both institutions and carries out cutting-edge multi-disciplinary research in different topics related to climate. This multidisciplinary character embraces the study of the Earth System, from the core to the stratosphere, in order to solve scientific problems in Earth Sciences within the framework of scientific collaborations with other research organizations and industry, at both national and international levels.